



Residual Channel-attention (RCA) network for remote sensing image scene classification

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Abstract

High-resolution remote sensing (HRRS) image scene classification has gained increasing importance in recent years, with convolutional neural networks (CNNs) showing particular promise due to their proficiency in extracting spatial features. However, traditional CNNs face significant limitations. Specifically, they struggle to capture complex semantic relationships between objects at varying scales, and they lack the ability to effectively capture long-distance dependencies between features. This limitation is especially problematic in HRRS images, where spatial relationships and semantic content are deeply intertwined. Additionally, traditional CNNs are limited in handling substantial intra-class variation and inter-class similarity, which are common in remote sensing images. To overcome these challenges, we introduce a novel Residual Channel-attention (RCA) network for scene classification. The RCA network introduces a lightweight residual structure to better capture multi-scale spatial features and incorporates a channel attention mechanism that selectively emphasizes relevant feature channels while suppressing irrelevant ones. To further refine the focus on critical image features, we integrate a squeeze-and-excitation (SE) mechanism as a self-attention component, which helps the network prioritize the most informative features and ignore background noise. We evaluated the RCA network on three public datasets: RSSCN7, PatternNet, and EuroSAT, achieving classification accuracies of 97%, 99%, and 96%, respectively. The results demonstrate that superior of the RCA network compared to state-of-the-art strategies in remote sensing image classification. Furthermore, visualization using the Grad-CAM++ algorithm highlights the effectiveness of our channel attention mechanism and underscores the RCA network's robust feature representation capabilities.

Keywords Remote sensing · Object detection · Image classification · Attention mechanism · Deep learning

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1 Introduction

Remote sensing is a broad detection strategy that uses sensors to capture objects' electromagnetic wave properties and extract the object characteristics with their change regulations using data processing and analysis [1, 2]. Recently, remote sensing images have improved temporal, spatial, and spectral resolutions with the advancements in computer science and the expanding number of remote-sensing satellites. While traditional "object-based" and "pixel-based" classifications consider low-level features, the escalating quality and quantity of remote-sensing data necessitate an exploration of higher-level contextual information within images. The improvements in spatial, spectral, and temporal resolutions have resulted in a greater need to extract complex, high-level contextual information, as traditional methods relying on handcrafted features are unable to handle the increased complexity of remote sensing data. These advances have led to a shift toward deep learning methods, particularly CNNs, which can automatically learn abstract features and capture multi-scale, multi-dimensional relationships across different resolutions, making them more effective in processing HRRS data.

Image classification based on remote sensing is a fundamental and vital part of remote sensing image processing and analysis by utilizing the upper-level semantic information in the images, and it is widely used in land resource management [3, 4], military investigation, change detection, object detection [5–8], environmental monitoring [9–12], urban planning, and other fields. Since remote sensing images' highly complex spatial pattern and geometric structure, classification is severely hampered by redundant background. Furthermore, the variety of items makes classification more challenging. However, the intricate spatial patterns and geometric structures inherent in remote sensing images pose challenges to classification due to redundant backgrounds and a myriad of objects, contributing to significant intra-class variability and limited inter-class variety. Due to variances in optical properties such as geometry, size, texture, and orientation of the ground objects, scenes in the same semantic category frequently exhibit significant visual disparities in remote-sensing-based image classifications.

Different imaging circumstances can result in significant radiation intensity and colour changes in scenes of the same kind. Various scenes may have the same local object, whereas the same scene may include different local objects. As a result, scene classification would solve the dilemma of huge intra-class variability and minimal inter-class variety. Feature extraction is an important stage in image classification, and much of the present research focuses on defining image features more accurately. Traditional classification approaches rely on mid-level or low-level features [13]; however, these features are difficult to adequately describe the semantic content of remote sensing scenes, resulting in disappointing results. As a result, subsequent studies have concentrated on developing more complex approaches for automatically learning the high-level image features representations.

Deep learning has been developed in many computer vision applications, including object detection, segmentation, and scene recognition [14]. Researchers use the CNN networks in RS image classification to extract high-level semantic features. Most of them depend on the pre-trained CNN models for feature extraction in remote sensing images, such as VGGNet, CaffeNet [15], and GoogleNet [16]. In [17], the authors first used the pre-trained VGGNet for feature extraction and then performed discriminant correlation analysis to refine the extracted features from VGGNet, resulting in high classification performance. ResNet [18] is another popular convolutional neural network strategy used for image scene classification in remote sensing. ResNet is more comprehensive and deeper than VGGNet and can

successfully address network degradation using the residual structure. One notable deep learning architecture, XcelNet17, comprises fourteen convolutional and three fully connected layers, achieving good classification accuracy across five benchmark datasets. This model outperforms established networks such as AlexNet, VGG16, and ResNet50.

In this work, remote sensing images refers to the data captured by remote sensors, while scenes denote specific areas or contexts within these images that are subject to classification or analysis. Although these terms are related, they are not used interchangeably; scenes are a subset of the broader remote sensing images, representing the areas of interest for scene classification. Figure 1 shows how the main object generally occupies the majority of the image in the case of a natural image, while objects in remote sensing images are often tiny and scattered, and the redundant background takes up the majority of the image as shown in Fig. 1 (c). Furthermore, not all of the information in the image is relevant to recognizing the visual scene. Vehicles and roads are particularly significant for identifying highway scenes, as illustrated in Fig. 1 (b), while houses and trees in the background deteriorate the classification decision. As a result, when focusing on the features of remote sensing images, we should pay greater attention to the most discriminative features of objects and eliminate the background interference. Hence, we integrate the residual convolutional network with a channel attention strategy. That focuses on the most representative and discriminative image features, resulting in a computation reduction and performance improvement. Furthermore, as demonstrated in Fig. 1 (b), the considerable intra-class variability of remote sensing picture scenes makes classification extremely difficult. Depending on the location, season, or sensor, images in the same category might have a lot of diversity.

Generally, deep learning-based strategies can learn sophisticated semantic features and more abstract images and more accurately recognize and describe image scenes, resulting in significant improvements in classification accuracy. The use of deep learning in image processing has recently experienced considerable growth but still has challenging issues for remote sensing image classification because such methods are mainly concerned with extracting global features while ignoring the crucial objects information in the image. Addressing these challenges requires a shift from traditional mid-level or low-level feature reliance to more sophisticated approaches. In this context, deep learning, particularly convolutional neural networks (CNNs), has emerged as a powerful tool for extracting high-level semantic features from remote sensing images. Noteworthy CNN architectures, including VGGNet,

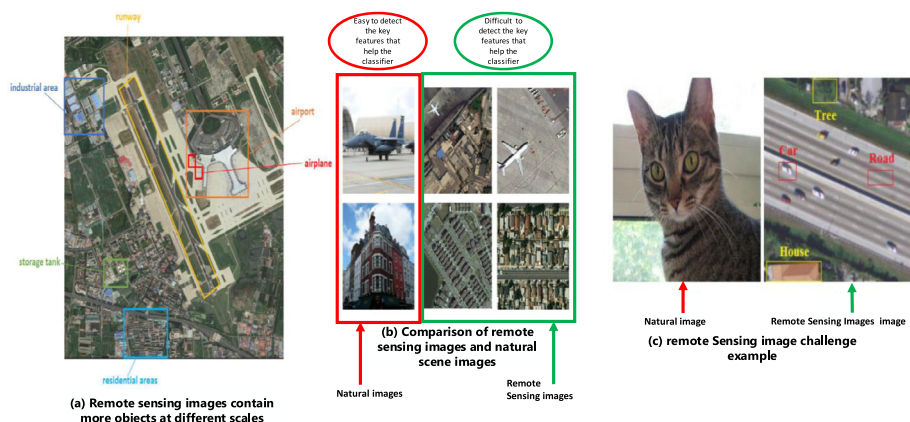


Fig. 1 Comparison of natural images and remote-sensing images

CaffeNet, GoogleNet, and ResNet, have demonstrated their efficacy. Despite their success, challenges persist, as CNNs often focus on extracting global features, neglecting crucial object information within images. To bridge this gap, we propose the Residual Channel-attention network (RCA), a novel method that integrates a lightweight residual network for spatial feature extraction and a channel attention mechanism to emphasize important feature channels selectively. The incorporation of a squeeze-and-excitation network as a self-attention mechanism enables a focus on the most informative features related to remote sensing images, disregarding irrelevant information. In summary, this study provides the following four main contributions:

- We propose a self-attention deep network structure for image scene classifications based on remote sensing that incorporates the channel-attention mechanism into the residual convolutional network. The self-attention mechanism drives the network to focus on the most critical parts of images while suppressing background disturbance. It can significantly reduce the processing time and enhance classification performance, especially for tiny datasets or images with complicated backgrounds.
- A new architecture of Residual connections: the residual connections in ResNets with SE attention allow the network to pass the input data through multiple layers without the risk of vanishing gradients, making it easier for the network to learn complex relationships between the input and output data.
- End-to-end learning: With the combination of residual connections and channel attention, the ResNet with channel attention is able to learn end-to-end from the input data to the final prediction. This makes the network more flexible and capable of adapting to different input distributions and learning more complex representations of the data.
- The effectiveness of the proposed model and its ability to achieve better image classification results compared with previous state-of-the-art approaches has been demonstrated through extensive experiments on three benchmark datasets.

The rest of this work is organized as follows. Section 2 summarizes the related work. Section 3 presents the proposed strategy. Sections 4 and 5 reports the experimental results and discussion. Section 6 demonstrates the conclusion.

2 Related works

The remote sensing image scene classification strategies can be divided into three groups based on the extracted features' semantic levels: high-level features, mid-level features, and low-level features. The low-level feature methods aim to represent an image by extracting its fundamental properties before identifying the scene. So, feature vectors generated from low-level visual qualities are commonly used to describe remote sensing image scenes, divided into global and local feature vectors. Local feature descriptors, such as the histogram of oriented gradients (HOG) and the scale-invariant feature transform (SIFT), are commonly employed to represent the local structures of remote sensing pictures for conveying complicated structural information. While global feature descriptors such as texture feature and colour histogram are commonly used to characterize the spatial arrangements of images. However, When the scene's structure and spatial arrangements are uniformly distributed, methods based on low-level features can yield good results, but they are limited in complicated scenarios.

Mid-level features methods aim to learn a set of fundamental functions used for feature encoding, such as the bag of visual words (BoVW) mode. However, such approaches have

drawbacks, including severe quantization errors that might result in the loss of crucial information. As a result, many feature coding approaches have been suggested to make better use of spatial information, including spatial pyramid matching (SPM), probabilistic latent semantic analysis (pLSA), and improved Fisher kernel (IFK). Both low-level and medium-level features-based techniques rely mostly on hand-crafted features, limiting their versatility and applicability to a wide range of applications.

High-level feature-based strategies such as deep learning have recently received more attention in computer vision, and major advances in remote sensing image classification based on deep features have been accomplished. Deep learning approaches outperform strategies based on low-level and mid-level features since the convolutional neural network (CNN) can automatically acquire more abstract and discriminative semantic information. For feature abstraction, AlexNet employs a large number of convolutional kernels. VGGNet makes advantage of deeper and larger networks to improve the network's feature learning skills. In comparison to VGGNet and AlexNet, GoogleNet can extract feature maps of many perceptual domains, improving the width and depth of the network while lowering the network parameters. ResNet uses the residual structure to tackle the network degradation problem when the model is deeper. DenseNet employs a densely linked topology that allows features to be reused and the number of network parameters to be reduced. However, the efficiency of employing such basic structures in scene classification is restricted due to the massive scale and sophisticated content of image classification in remote sensing, as well as their features of minimum inter-class variety and tremendous inter-class variability.

Authors in [19] introduced a new image classification strategy based on spatial scale CNN blocks that help in extracting and adaptively adjusting abundant spatial features to enhance the informative features function in the classification results. In [20], the authors proposed a bilinear CNN-based feature extraction structure to improve the classification accuracy, and they used a depthwise separable convolution to solve the computation complexity and volume. A lightweight deep network is suggested in [21] for remote sensing images by incorporating the benefits of multidilation pooling and the dilated convolution in MobileNet V2 network, extracting multiscale features and improving the classification accuracy. In [22], the authors focus on the image texture features instead of the single CNN layer features, and they presented a new coding strategy to extract each convolutional layer information, including global, local, bottom, and deep. Authors in [23] extract both local & global features from the spatial domain and rotation invariant features from the frequency domain, using a new joint learning module including CNN and bandpass filtering network.

CNN-based algorithms have received massive interest in recent years, and their feature maps can reflect the scenes and achieve good classification performance [24]. However, there is little emphasis on the important regions with their features, and the redundant regions cause classification problems. Researchers have utilized various feature improvement techniques to overcome this issue, including identifying more effective means of feature fusion [17], selecting more efficient feature maps [25], and incorporating attention processes into CNN [26].

Recent developments in image classification for remote sensing applications have witnessed a rise in innovative strategies to cope with the challenges posed by complex spatial patterns and diverse semantic content [27, 28]. One notable avenue of exploration involves the integration of graph convolutional networks (GCNs) to enhance contextual modeling. GCNs leverage graph structures to capture relationships between pixels, thereby providing a more holistic understanding of scene elements and their interplay. This approach proves particularly valuable in scenarios where spatial dependencies are critical, contributing to improved accuracy in classifying remote sensing images with intricate object relationships.

Authors in [29], propose dual Graph U-Nets for Hyperspectral imaging (HSI) classification (DGU-HSI) addresses limitations in conventional graph convolutional neural networks by integrating spatial and spectral graphs for feature extraction. While most models only consider spectral information for graph construction, DGU-HSI introduces a dual GCN that simultaneously extracts spatial and spectral features.

In [30], a multi-branch deep learning framework is introduced for land scene classification, addressing challenges such as complex textures, cluttered backgrounds, small object sizes, and scale variations. The proposed framework efficiently integrates global contextual features and multi-scale features through two branches, where the first extracts global contextual information, and the second employs a fully convolutional network (FCN) for multi-scale local feature extraction. These methods demonstrate good classification performance over existing CNN-based state-of-the-art methods, but the computational complexity and model complexity are still challenges that need to be overcome.

In recent years, a slew of studies based on attention mechanisms have developed in the field of remote sensing image classification. Authors in [31] proposed a fuzzy classification approach using a multi-scale strategy to extract visual attention features of high-resolution remote sensing images. To enhance the classification performance, authors in [32] construct a recurring attention model which deceptively picks certain crucial regions in images while discarding some non-critical information. In order to utilize both the local and global information in aerial image classification, a new end-to-end global-local attention network is presented in [33]. The local semantic information and the global information are learned by replacing the fully connected layers for VGGNet by a global and local attention branches. In [34], the authors overcome the problems associated with CNN and the visual attention strategies such as extracting information from the whole scene and the problem of repeated texture and intro-class diversity, respectively. A new attention deep feature framework is presented, which generates the attention map with multiplicative fusion as a clear input to drive the network focusing on discriminating regions and enhancing the classification accuracy.

Authors in [35] adopt the self-attention with a generative adversarial network in which all features locations are exploited to overcome the problem of extracting local spatial features with a lower-resolution map. Moreover, they used spectral normalization for the generative adversarial network to enhance the training dynamics. A new object classification architecture is introduced in [36] by combining Capsule Network and a Self-Attention strategy. A shallow CapsNet network with self-attention help is used instead of the deeper network to extract important features with low computational complexity. Self-attention strategy obtained a better accuracy with the easy train when dealing with fewer data samples and images with complicated backgrounds since it can help the network to focus on more significant portions of the image. However, more investigation of both the convolution neural network and the attention needs to be implemented to achieve better accuracy.

3 The proposed algorithm

This paper is motivated by the potential to enhance scene classification performance in remote sensing by employing two types of networks. Specifically, while the channel attention network possesses dynamic global receptive fields, it lacks the desired inductive bias. On the other hand, convolution neural network exhibit strong inductive biases but are limited by their deficiency in large receptive fields. Thus, it remains unclear whether fusing these two networks can yield improved performance in scene classification. In order to address this

issue, we have proposed a new architecture called RCA by combining the strengths of both residual and channel attention, as shown in Fig. 2.

The proposed Residual Channel-attention network (RCA) introduces a novel approach to scene classification in high-resolution remote sensing images, addressing the specific limitations of traditional convolutional neural networks (CNNs). Traditional CNNs are effective at extracting spatial features but often struggle to capture complex semantic relationships between objects at varying scales. Moreover, CNNs typically focus on global features and may overlook long-distance dependencies and critical local features within the image, which are essential for accurately classifying remote sensing scenes that contain diverse objects. To overcome these limitations, we combine a lightweight residual structure with a channel attention mechanism, specifically the squeeze-and-excitation (SE) network, into the proposed RCA network.

The motivation behind combining the channel attention mechanism with CNNs is to leverage the strengths of each approach: CNNs excel at extracting hierarchical spatial features, while channel attention mechanisms, such as the SE network, enhance the model's ability to focus on the most relevant feature channels, which traditional CNNs tend to ignore. The residual connections in the RCA network ensure effective multi-scale feature extraction, allowing the network to handle both fine and coarse details without the degradation of learning performance. This helps mitigate issues such as vanishing gradients, which are common in deep networks, enabling the network to learn more complex relationships between input and output data. The SE network plays a pivotal role by adaptively recalibrating the feature responses across channels. Through the squeeze operation, it aggregates global information across the entire spatial domain of the feature map. Then, during the excitation phase, it selectively emphasizes important feature channels while suppressing irrelevant ones. This recalibration ensures that the network allocates more attention to the most informative channels, thereby improving the overall feature extraction process. This mechanism helps address the limitation of CNNs that often fail to prioritize key features crucial for remote sensing image classification.

The integration of the SE network within the residual structure of the RCA network is thus a strategic choice that combines the spatial feature extraction capabilities of CNNs with the adaptive focus provided by channel attention. The collaboration between the residual connections and the SE mechanism results in enhanced spatial and channel-wise feature extraction. The residual connections allow the model to learn features across multiple layers efficiently, capturing the intricate spatial patterns of remote sensing images. Meanwhile, the SE network

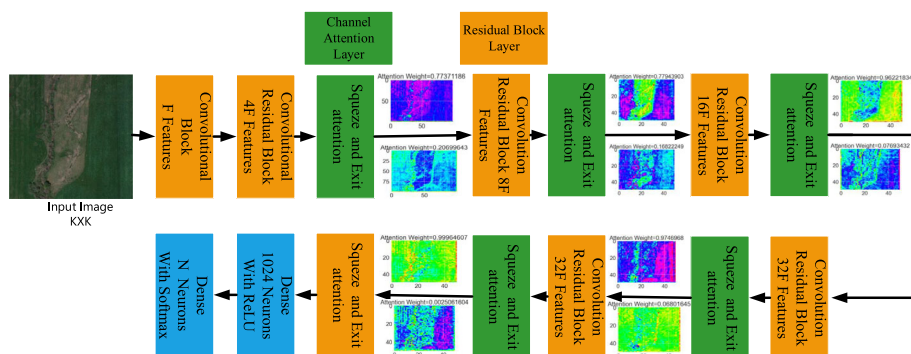


Fig. 2 Workflow chart of the proposed system including the residual block and channel attention layers

selectively emphasizes spatial features relevant to scene classification, overcoming CNNs' inherent limitation of neglecting fine-grained, channel-specific information. Together, these mechanisms ensure that the RCA network is more robust and effective in capturing both local and global dependencies, making it superior to traditional CNN-based approaches in remote sensing image classification.

The proposed RCA network introduces several key innovations for improving remote sensing image classification. These contributions are as follows:

- **Self-Attention-Based Deep Network:** We introduce a novel self-attention deep network structure that incorporates a channel-attention mechanism within a residual convolutional network. This mechanism enables the model to focus on the most critical image regions while reducing background interference, resulting in improved classification performance, particularly for images with complex backgrounds.
- **Enhanced Feature Representation using Residual Connections:** The residual connections in the RCA network allow for effective multi-scale feature extraction without the risk of gradient vanishing. This structure improves the network's ability to capture both local and global spatial features, enhancing its capacity to learn complex relationships within the data.
- **Lightweight Channel-Attention Mechanism:** The integration of a lightweight squeeze-and-excitation (SE) channel-attention mechanism allows the RCA network to selectively emphasize important feature channels while suppressing irrelevant ones. This helps the model focus on key features that are crucial for accurate scene classification.

Our approach involves extracting low-level, local features such as corners and edges using convolution operations in the network's shallower layers, followed by capturing global abstract semantic features using channel-attention module in the deeper layers. The last step involves inputting these distinguishing characteristics into the fully connected layer to predict the remote sensing scene category. The RCA network, as shown in Fig. 2, comprises five convolutional residual blocks, five channel attention blocks, an adaptive average pooling layer, and a fully connected layer. The main steps of the proposed scheme are described in the following subsections.

3.1 Spatial feature detection backbone

This part is employed to extract features from various scales, and what's more, the residual connected structure interlinks the features from these various scales, resulting in better performance compared to conventional CNNs in representing the intricate semantic connections among multiscale objects in remote sensing images. CNNs have demonstrated exceptional abilities in representing features for scene classification tasks. Nevertheless, remote sensing scene datasets typically contain a limited number of images, and conventional CNN approaches often have a large number of parameters and shallow network layers. Overfitting is commonly caused by limited training data, while shallow network layers may not extract enough high-level information. Nonetheless, deep networks face issues such as gradient disappearance, which can compromise the benefits of increasing the network depth.

In order to address this issue, the ResNet network is utilized in this paper as the foundational network, which was introduced in 2015 and is based on a residual structure as its core module. The residual learning unit, as depicted in Fig. 3, consists of multiple convolutional layers and skip connections. It effectively transforms the input features into output features while

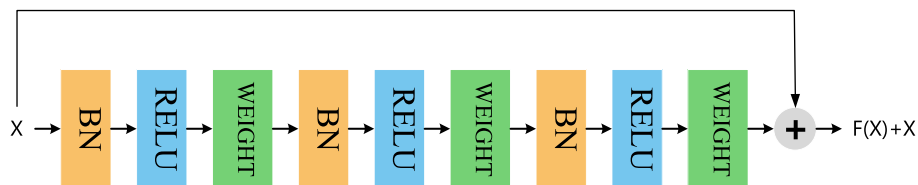


Fig. 3 Residual convolutional block

maintaining consistency. The following are the definitions of its components.

$$\gamma = \sigma(F(X) + X) \quad (1)$$

$$F(X) = W_2\sigma(W_1X) \quad (2)$$

where γ is the output, and X is the input. $F(X)$ represent the residual mapping that must be acquired during the learning process, σ is the correction unit, W_1 and W_2 are the weights for each layer, and $F(X)+X$ is the output of the residual module.

The use of residual structures in deep learning has revolutionized the traditional approach by allowing direct mapping of input data to output while maintaining a constant residual connection. This approach prevents the loss of information during the transfer process. In the context of scene classification, when the difference between image categories is minimal, the model requires in-depth learning of image features to achieve accurate classification. However, when the difference between image categories is significant, shallow features alone may suffice for classification, rendering deeper features unnecessary. In such situations, residual modules can preserve essential features in the lower layers, preventing network degradation and improving model performance.

3.2 Channel feature attention layer

In light of Residual's robust feature extraction capabilities within the spatial domain, an attention mechanism known as "squeeze-and-excitation blocks," was incorporated into the feature channel domain. This attention mechanism operates by modifying the weights of distinct feature maps within the channel domain to selectively choose significant features. In order to leverage the channel attention module to its full potential without introducing a large number of parameters, the channel attention mechanism is exclusively applied to the dense block and transition layers. Additionally, the channel attention network is designed to be compact and efficient, contributing only 0.22M parameters and posing no risk of overfitting. The transition layer comprises a 1×1 convolution layer and an average pooling operation with a stride of 2 that works to decrease the size of the feature map. An adaptive downsample mechanism, referred to as the channel attention module, is incorporated into this transition layer. Figure 4 illustrates that the channel attention mechanism processes feature channels in two stages: "squeeze" and "excitation."

Our proposal involves utilizing the squeeze-and-excitation network following each residual block to facilitate the learning of interdependencies among various feature maps. During the squeeze stage, the input features undergo compression into a one-dimensional (1-D) vector with a length equivalent to the number of channels. The initial input feature has a size of $H \times W \times C$, where $H \times W$ represents the spatial domain size and C is the number of channels. The compression of each spatial domain $H \times W$ is achieved through global average pooling, which yields a value output. The final output of the squeeze stage is a vector of size $1 \times 1 \times C$.

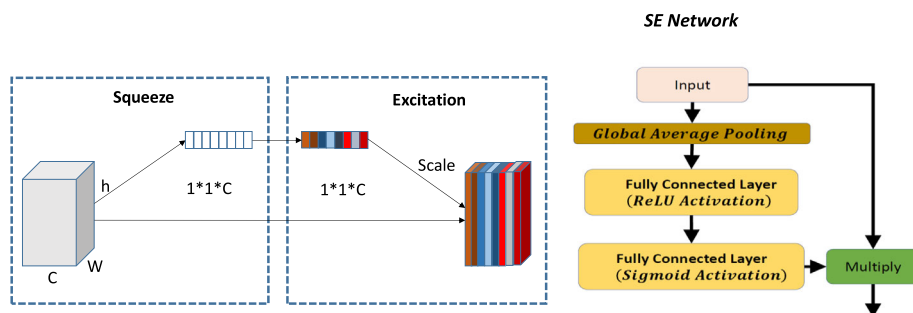


Fig. 4 Channel attention model

C. More specifically, the calculation of the c th element of the squeeze output, denoted as S_q , is expressed as follows:

$$S_q(NO, Z) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W D_0(NO, i, j, Z) \quad (3)$$

where, $H \times W$ represents the dimension of the output (Do) of the residual block. The squeeze output S_q is derived from this output. Furthermore, NO and Z correspond to the batch size and the number of feature maps, respectively. The excitation stage involves capturing interdependencies among channels via a gate mechanism that comprises two nonlinear fully connected layers. The resulting output of the excitation stage is denoted as E , and is computed using the formula

$$E = \delta(W_2 \alpha(W_1 S_q)), \quad (4)$$

where W_1 and W_2 represent the weight matrices of the fully connected layers, α denotes the *ReLU* activation function, and δ refers to the sigmoid activation function. The sigmoid activation function is represented by δ in the *SE* network, which leverages two fully connected layers featuring the *ReLU* and *sigmoid* activation functions, respectively. W_1 and W_2 denote the weight matrices of the first and second fully connected layers, respectively. The output of the *SE* network is obtained by multiplying the input of the network D_0 with the excitation output E . This can be mathematically represented as:

$$SE = D_0 \circ E \quad (5)$$

Here, the symbol $\circ$ denotes element-wise multiplication. The primary objective of the *SE* network is to scale the feature maps and aid the network in identifying significant features from the input data, which are highly relevant to the final classification outcome.

The process of recalibrating or highlighting each channel in accordance with its importance aims to create a more robust representation by enhancing the crucial features. This is achieved by assigning varying levels of importance to different channels, based on their relevance to the final classification outcome.

The channel attention module facilitates the assignment of adaptive weights to different features through the process of squeezing and expanding the feature channels. This approach results in fewer parameters when compared to the attention model for feature maps, thereby reducing the potential risk of overfitting.

4 Experimental results

The proposed RCA network's performance was assessed through a set of experiments utilizing three public remote sensing scene image datasets: RSSCN7, PatternNet, and EuroSAT. The experimental configurations, outcomes, and a thorough analysis of the findings are comprehensively presented below.

4.1 Dataset description

4.1.1 RSSCN7 dataset

The RSSCN7 dataset was publicly released in 2015 by Wuhan University, containing a total of 2800 images across seven distinct scene categories: industry, grass, residential, field, forest, lake, parking lot, and river. Each category in the dataset consists of 400 images, sampled at one of four different ratios (1:1300, 1:700, 1:5200, and 1:2600), with 100 images corresponding to each ratio. The images are all 400 x 400 pixels in size and were acquired in various seasons and weather conditions. The varying sampling ratios present a challenge for classification of the dataset. Figure 5 displays the different image categories.

4.1.2 PatternNet dataset

PatternNet is a remote sensing dataset that includes 38 distinct classes, each consisting of 800 images that are 256 x 256 pixels in size. The images in PatternNet were sourced from Google Earth imagery or obtained via the Google Map API for specific US cities. Figure 6 displays the different image categories.

4.1.3 Eurosat dataset

The EuroSAT dataset covers 13 spectral bands and consists of 10 distinct classes. This dataset contains 27,000 images, with each of the 10 classes consisting of 2,000 to 3,000 images. The images are in the form of patches, with each patch measuring 64 x 64 pixels. Figure 7 displays the different image categories.

4.2 Accuracy assesmnet

In order to effectively compare the performance of our proposed method to that of other experimental methods [13, 37–44], we opted to utilize the same training-to-test set ratios that were employed in the majority of prior research. Specifically, we set the training ratios to 80% for the PatternNet, Eurosat, and RSSCN7 datasets. This involved randomly selecting



Fig. 5 Snapshot images for RSSCN7 dataset: (a) grass; (b) field; (c) industry; (d) river lake; (e) forest; (f) resident; (g) parking



Fig. 6 Snapshot images for types in the patternNet dataset

images from each dataset to be used for training, with the remaining images being used for testing. To ensure the accuracy and reliability of our experimental results, we repeated the experiment ten times for each of the three datasets, randomly dividing them according to the training ratios. The final results of the proposed network were calculated as the mean and standard deviation of these experiments.

In order to assess the efficacy of our proposed model, we utilize both the confusion matrix and overall accuracy measures. Through the use of the confusion matrix, discrepancies between predicted and true labels can be easily identified.

Overall accuracy, on the other hand, is a measure of the ratio of correctly predicted values to the total number of predicted values across all test sets. This metric provides an evaluation of the model's performance in accurately classifying data.

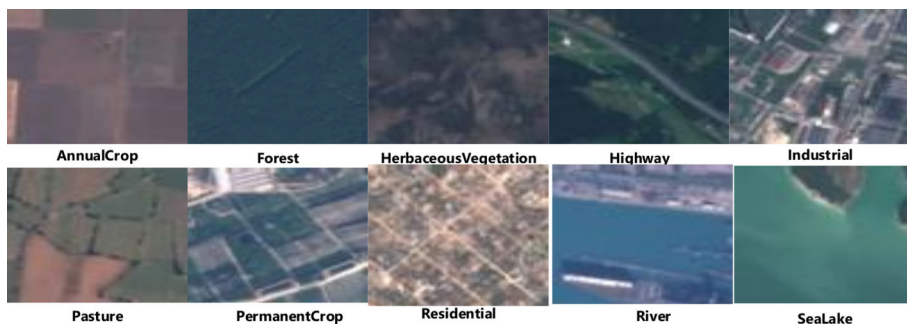


Fig. 7 Snapshot images for types in the Eurosat dataset

4.3 Experimental details

The proposed model discussed in this article has been implemented using the Tensorflow framework. During our experimentation, we set the training epoch to 500 and each epoch has a batch size of 4. The parameter optimizer we used for the model is Adam, and the initial learning rate was fixed at 0.001. In order to improve the quality of the model, data augmentation was performed during the training process. The trade-off parameters of the cross-entropy loss and the center loss were set to 1 and 0.008, respectively. The equations for cross-entropy loss, center loss, and total Loss are given by:

$$L_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M y_{ij} \log(\hat{y}_{ij}) \quad (6)$$

Where:

- N is the number of training samples.
- M is the number of classes.
- y_{ij} is a binary indicator (0 or 1) if class label j is the correct classification for input i .
- $\hat{y}_{ij}$ is the predicted probability of input i belonging to class j .

$$L_{center} = \frac{1}{2} \sum_{i=1}^N \|f(x_i) - c_{y_i}\|_2^2 \quad (7)$$

Where:

- $f(x_i)$ is the feature representation of sample i .
- c_{y_i} is the center of the feature representations of class y_i .
- $\|\cdot\|_2$ denotes the Euclidean distance.

$$L = L_{CE} + \lambda L_{center} \quad (8)$$

Where λ is the trade-off parameter between the cross-entropy loss and the center loss.

Our experiments were carried out on a GPU with the following configuration: Intel(R) Xeon(R) E5-2670 v3 CPU, 64GB memory, GeForce GTX TITAN X, and 12GB of display memory. In our experiment, we conducted training and testing of models using three different datasets: PatternNet, Eurosat, and RSSCN7. To facilitate this, we divided each dataset into two parts - a training set and a test set. Given the variation in the number of images across the different datasets, we randomly generated the training and test sets for each dataset.

In order to verify the effectiveness of the proposed model, we conducted a comparison with several existing remote sensing image scene classification methods. These include, CaffeNet [13], GoogleNet [13], VGG16 [13], DBN [37] GCFs+LOFs [38], Deep Filter Banks [39], RSSCNet [40], GCNN [41], ETDL [43] pre-trained ResNet-34 [40], and MTLS [44]

4.4 Performance analysis

4.4.1 Experimental results on different dataset

This section tests the proposed strategy's efficacy on different datasets, including RSSCN7, PatternNet, and Eurosat datasets, to investigate its robustness and generalization. The performance comparison results for RSSCN7 dataset are presented in Table 1. According to

Table 1 The overall accuracy of different algorithms on RSSCN7 dataset

Method	Overall Accuracy %
DBN [37]	77
GoogleNet [13]	85
CaffNet [13]	88
VGG [13]	87
GCFs+LOFs [38]	95
Deep Filter Banks [39]	90
RSSCNet [40]	97
GCNN [41]	94
HFCC-Net [45]	95.30
EfficientNetB3-Attn-2 [46]	96.17
L2RCF [47]	96
Propose Method	97

the table, it can be observed that traditional algorithms, including DBN architecture has the poorest performance. In contrast, deep network-based models such as CaffeNet, GoogLeNet, and VGG16 have significantly improved the classification accuracy compared to traditional algorithms. Additionally, some enhanced deep learning-based methods have shown even better performance than these popular deep networks. Comparable classification accuracy is achieved by some other deep learning-based methods, such as EfficientNetB3-Attn-2, L2RCF, and RSSCNet. However, the proposed model outperforms all algorithms, exhibiting remarkable classification accuracy. When the training ratios are 80%, the proposed model achieves overall accuracies of 97%. These results indicate that the channel-attention mechanism in our model enhances the discrimination of features. The confusion matrix of the proposed model on the RSSCN7 dataset, with a training ratio of 80%, is depicted in Fig. 8. Our model attains 97% accuracy in 7 categories, and most categories obtain more than 95% accuracy. However, the “Industry” category has relatively lower accuracy, only 93%. This could be due to the similar spatial distribution of industry in the “industry” category, leading to the loss of classification accuracy by the interference of these similar features.

Table 2 shows the results of our proposed method compared with four state-of-the-art remote sensing image scene classification approaches on the PatternNet dataset. As clear from Table 2, the overall classification accuracy of our proposed method is 99% on the PatternNet test dataset, which is 7%, 4%, 4%, and 2% higher than that of multitask learning network (MTLN), error-tolerant deep learning (ETDL), Extracting Hierarchical and Collaborative intertarget feature Alignment (EHACA), and VGG-16, respectively. As shown in Fig. 9, it is asserted that the proposed strategy achieves satisfactory accuracy which attains 97% accuracy in 38 categories, and most categories obtain more than 98% accuracy. Table 3 shows the comparison results on the Eurosat dataset. As clear from Table 3, the overall classification accuracy of our proposed method is 96.3% on the Eurosat test dataset, which is 7.13%, 2.15%, 2.13%, and 1.55% higher than that of Pretrained Resnet-34 on ImageNet, Pretrained Resnet-34 on ImageNet with Finetunning, MTLN, and Multi-Branch Deep Learning Framework (MBDLF), respectively. As shown in Fig. 10, it is asserted that the proposed strategy achieves satisfactory accuracy which attains 96.3% accuracy in 10 categories, and most categories obtain more than 95% accuracy.

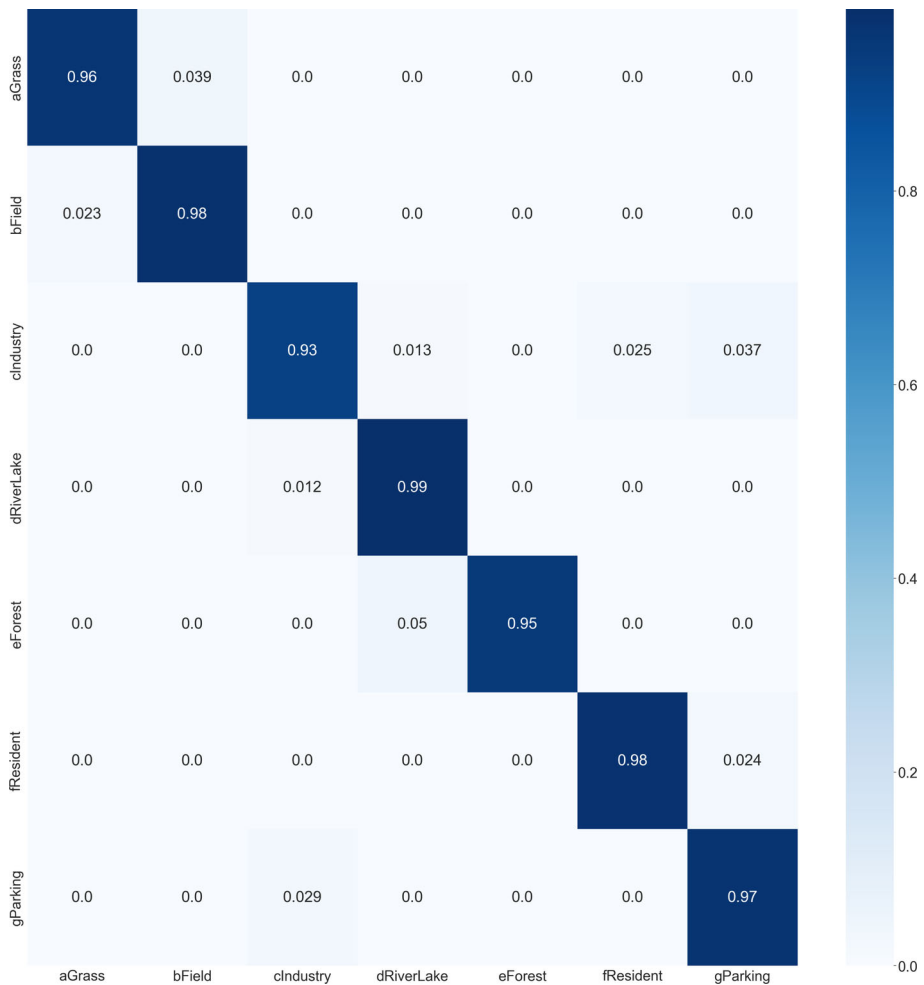


Fig. 8 Proposed method classification confusion matrix for RSSCN7 dataset

4.4.2 Features and attention maps visualization on the proposed RCA network

We aim to focus more on the key image channels, which contain the most distinguishable feature maps, that are crucial for scene categorization. Simultaneously, we aim to eliminate irrelevant and misleading channels.

Table 2 The overall accuracy of different algorithms on PatternNET dataset

Method	Overall Accuracy %
VGG16 [42]	97
ETDL [43]	95
MTLN [44]	92.5
EHACA [48]	95
Propose Method	99

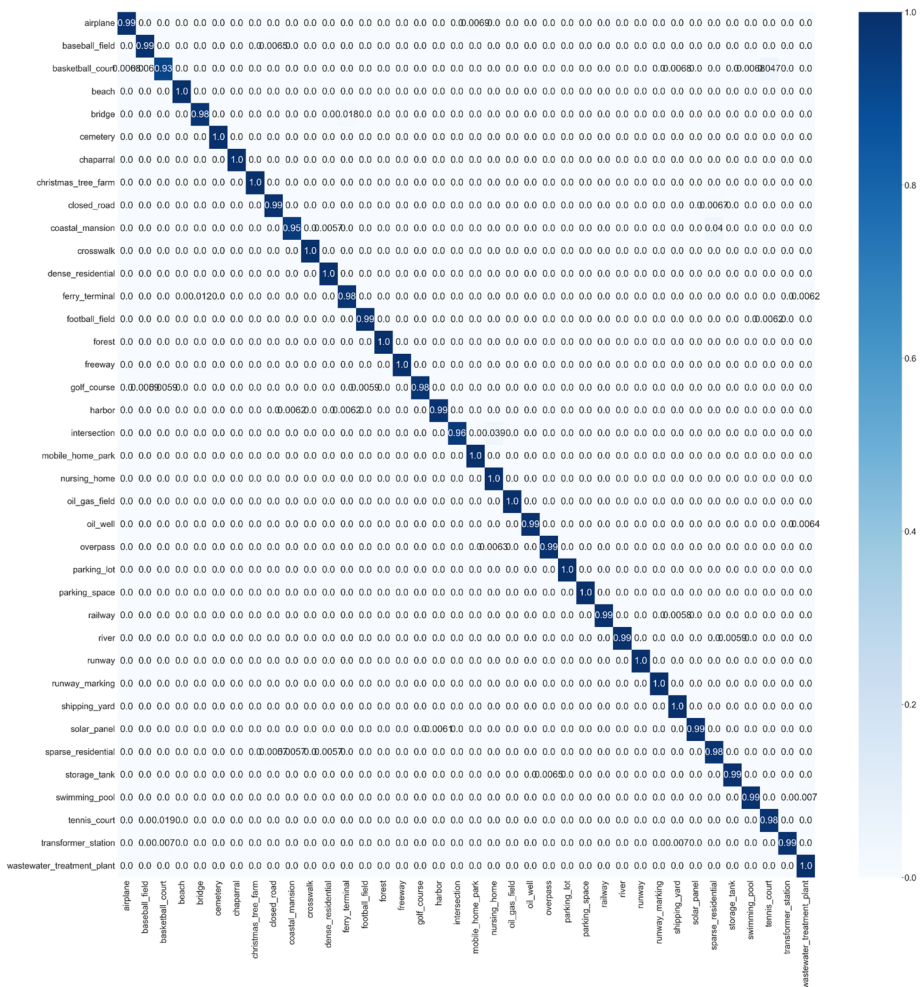


Fig. 9 Proposed method classification confusion matrix for PatternNET dataset

The performance with and without the SE mechanism is analyzed to demonstrate how it enhances feature representation. As shown in Fig. 11, the green box represents the channel feature map with the maximum weight while the red box represents the green box with

Table 3 The overall accuracy of different algorithms on Eurosat dataset

Method	Overall Accuracy %
Pretrained Resnet-34 on ImageNet [42]	89.17
Pretrained Resnet-34 on ImageNet+Fine tuning [42]	94.15
MTLN [44]	94.17
MBDLF [30]	94.75
Propose Method	96.3



Fig. 10 Proposed method classification confusion matrix for Eurosat dataset

the minimum weight. The maximum weight is associated with the channel features that effectively differentiate between the objects in the scene, while the minimum weight is associated with the channel features that cannot effectively differentiate between the objects in the scene. The results depicted in Fig. 11 indicate that the network which was augmented with channel attention can effectively extract essential information from these scenes. The primary objects present in each scene are precisely captured, which greatly assists in the classification of the scenes. Hence, channel-wise attention was implemented to readjust the weights of multi-scale features that were extracted using the residual network. The feature maps that have a direct impact on the ground-truth label are assigned the highest attention weights, while the feature maps that contain irrelevant or misleading information are given the lowest attention weights. These feature maps provide no additional guidance to the network for accurate classification.

Moreover, the Grad-CAM and Grad-CAM++ algorithms generated attention maps which are illustrated in the first row of the figures from Figs. 11, 12, 13 and 14. The regions in

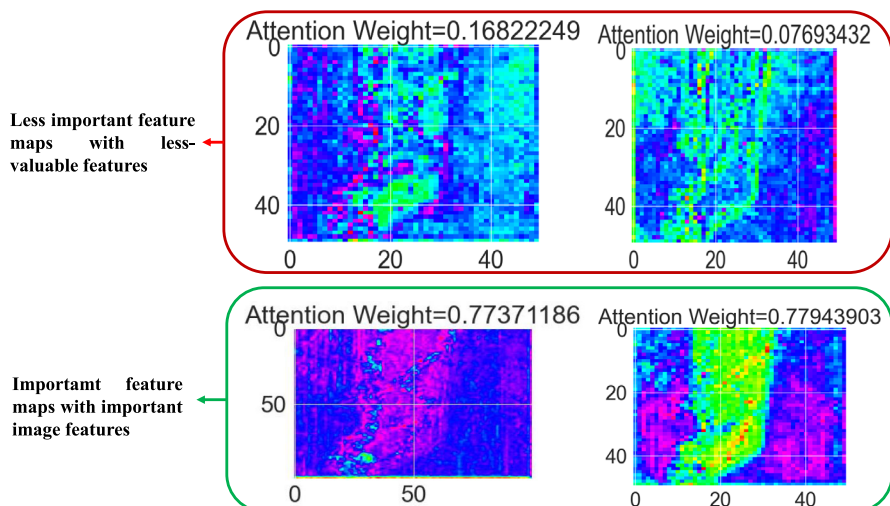


Fig. 11 Example from the Grass scene from RSSCN7 dataset. The first row represents example of lowest channel features after the channel attention layers that considered as a low important feature maps. The second row represents example of highest channel features after the channel attention layers that considered as an important feature maps

the feature map that are brighter indicate the regions that are more crucial for classification. Four different scenes were selected, namely a Grass, an airplane, and a storage tank. As shown in the first row of figures, the RCA network with channel attention achieves more precise localization of important areas for airplane, grass, and storage tank categories while

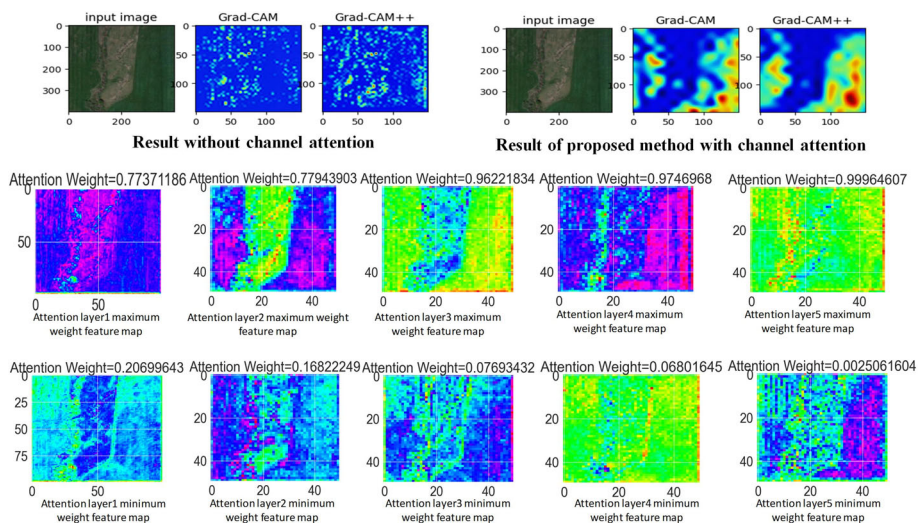


Fig. 12 Feature Visualization example for Grass scene in RSSCN7 dataset. First row: Original image, Grad-CAM and Grad-CAM++ Visualization without the channel attention, Second row: examples of maximum weights channel features for the five attention layers. Third row: examples of minimum weights channel features for the five attention layers

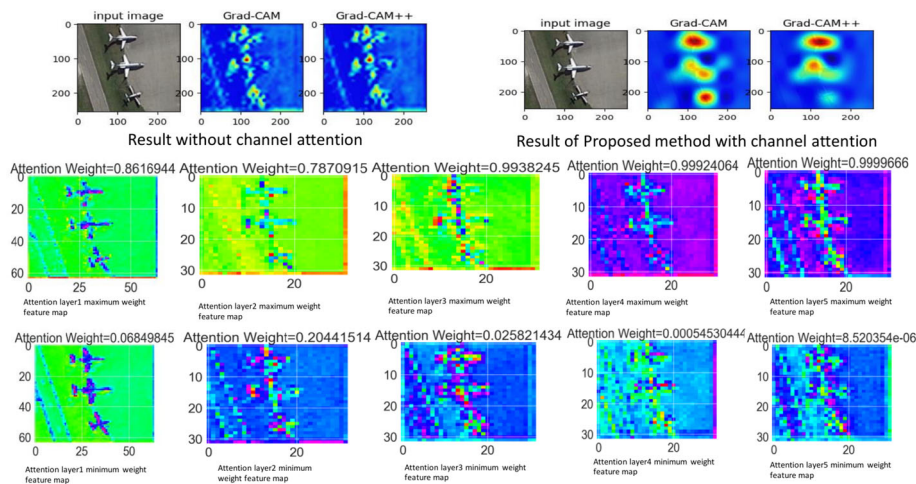


Fig. 13 Feature Visualization example for airplane scene in PatternNet dataset. First row: Original image, Grad-CAM and Grad-CAM++ Visualization without the channel attention and visualization with the channel attention, Second row: examples of maximum weights channel features for the five attention layers. Third row: examples of minimum weights channel features for the five attention layers

minimizing interference from low-priority features. Furthermore, despite the challenging task of distinguishing between the grass and non-grass regions, the proposed network including channel attention demonstrates improved control over crucial areas for these categories. These results indicate the effectiveness of the channel attention mechanism and the robust image feature representation capabilities of the RCA Network.

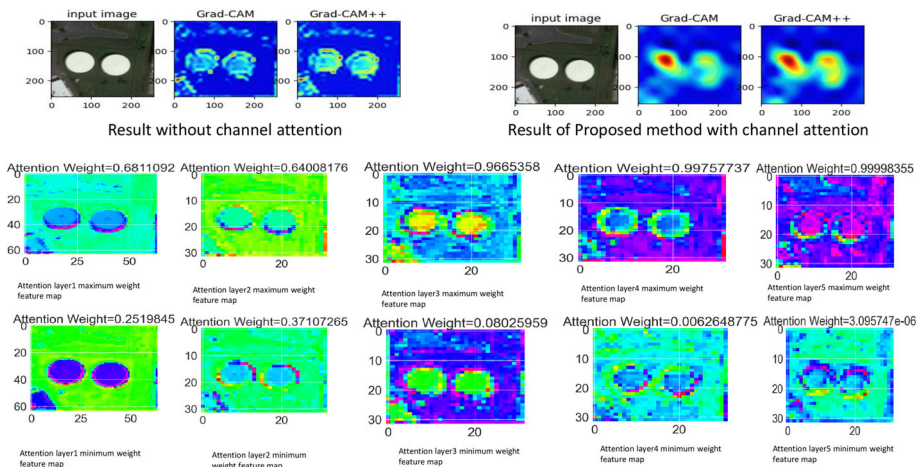


Fig. 14 Feature Visualization example for storage tank scene in PatternNet dataset. First row: Original image, Grad-CAM and Grad-CAM++ Visualization without the channel attention and visualization with the channel attention, Second row: examples of maximum weights channel features for the five attention layers. Third row: examples of minimum weights channel features for the five attention layers

5 Discussion

In this section, we delve into the outcomes derived from the experimental results presented in Tables 1–3. Based on these outcomes, it is evident that the RCA network demonstrates commendable performance across all three datasets. However, its efficacy varies, achieving optimal results with the PatternNet dataset, while showing comparatively lower performance on the RSSCN7 and EuroSAT datasets. The RCA network's superior performance on PatternNet can be ascribed to notable inter-class distinctions and substantial intra-class similarities. The clearly defined boundaries between classes in PatternNet enable the network to accurately capture unique features within each category. Additionally, the RCA model's channel-attention mechanism enhances the model's ability to identify variations in scale and orientation within the dataset, as well as to discern details at a granular level. These attributes are further reflected in the RCA network's high classification accuracy of 99% in Table 2, underscoring its capability to perform well on datasets where scenes exhibit high feature consistency and separation. Table 2 presents the outcomes of our proposed methodology in comparison with four cutting-edge approaches for remote sensing image scene classification using the PatternNet dataset. As evident in Table 2, our proposed method achieves an impressive overall classification accuracy of 99% on the PatternNet test dataset. This result is notably superior, surpassing the accuracy of multitask learning network (MTLN), error-tolerant deep learning (ETDL), Extracting Hierarchical and Collaborative intertarget feature Alignment (EHACA), and VGG-16 by 7%, 4%, 4%, and 2%, respectively. The effectiveness of our approach is further substantiated in Fig. 9, illustrating its commendable accuracy, reaching 97% accuracy across 38 categories, with most categories achieving over 98% accuracy.

In contrast, the framework exhibited diminished performance for the EuroSAT dataset in accurately identifying the herbaceous vegetation class. The network primarily encounters confusion between the herbaceous vegetation class and the permanent crop class, as both categories share analogous appearance and morphological features. These outcomes highlight the RCA network's ability to harness clear spatial distinctions within high-resolution datasets, underscoring its effectiveness in scenarios where class features are visually distinct. However, the Eurosat dataset aligns well with the RCA network's strengths due to its high-resolution imagery, distinct class boundaries, and a structured layout across the 10 categories. This dataset's lower inter-class similarity, especially among land-use classes such as built-up areas and forests, enables the RCA network to identify differentiating features with a high degree of accuracy. The experimental results, shown in Table 3 outlines the outcomes of our proposed method in contrast to four state-of-the-art approaches for remote sensing image scene classification, and demonstrates that our model achieved an overall classification accuracy of 96.3%, with most categories achieving over 95% accuracy. The overall classification accuracy of our proposed method stands at 96.3% on the Eurosat test dataset, marking notable improvements of 7.13%, 2.15%, 2.13%, and 1.55% over Pretrained Resnet-34 on ImageNet, Pretrained Resnet-34 on ImageNet with Finetunning, MTLN, and Multi-Branch Deep Learning Framework (MBDLF), respectively. Figure 10 visually reinforces the effectiveness of our proposed strategy, achieving a satisfactory accuracy of 96.3% across 10 categories, with most categories boasting over 95% accuracy. This investigation suggests that augmenting the network's depth with additional layers may not yield a substantial performance improvement, particularly in the context of aerial images.

It was additionally noted that deeper layers could prompt the network to discern features effectively in natural images, yet these deeper models exhibit suboptimal performance in aerial scenes. This observation is apparent in Table 3. The decreased performance on the EuroSAT dataset can be ascribed to its inclusion of more varied scenes and increased

inter-class similarities, particularly among the herbaceous classes, annual crop, and permanent crop. These challenges hinder the models from effectively learning differentiating features. Despite showcasing commendable performance on demanding datasets, the framework is not immune to limitations. From the results of our experiments, it is apparent that the network encounters challenges in distinguishing between two closely resembling images representing different classes. The inter-class similarity across various categories prompts the network to learn features that are too similar, leading to misclassifications within the framework.

Similarly, RSSCN7 presents unique challenges due to its higher inter-class similarity, varied backgrounds, and complex scene compositions, especially across natural landscapes. This dataset includes categories with overlapping visual characteristics, making it more challenging for the RCA network to differentiate between classes. The RSSCN7 dataset also features considerable scale and orientation variance within scenes, further straining the model's discriminative capacity. For instance, scenes of industrial and urban landscapes in RSSCN7 often resemble each other in spatial patterns, which contributes to a slight decline in accuracy due to the increased potential for feature overlap and misclassification. As evident from Table 1, traditional algorithms, including the DBN architecture, exhibit the least favorable performance. Conversely, deep network-based models such as CaffeNet, GoogLeNet, and VGG16 demonstrate significant improvements in classification accuracy compared to their traditional counterparts. Comparable classification accuracy is achieved by other deep learning-based methods like EfficientNetB3-Attn-2, L2RCF, and RSSCNet. However, our proposed model surpasses all algorithms, showcasing outstanding classification accuracy. With training ratios set at 80%, our model achieves an overall accuracy of 97%. These outcomes highlight that the channel-attention mechanism integrated into our model enhances feature discrimination. Our model attains 97% accuracy across 7 categories, with most categories achieving over 95% accuracy. Notably, the "Industry" category exhibits relatively lower accuracy, reaching only 93%. This discrepancy may be attributed to the similar spatial distribution of industry features within the category, resulting in reduced classification accuracy due to interference from these similar features. While the RCA model's architecture achieves effective classification on diverse datasets, RSSCN7's inherent complexity underscores the need for further optimization or additional contextual modeling to enhance feature discrimination across closely resembling classes.

Our analysis suggests that certain dataset characteristics, such as the level of inter-class similarity and the visual complexity of scenes, significantly impact the RCA network's performance. The differences in results across PatternNet, EuroSAT, and RSSCN7 reveal the RCA network's sensitivity to clear spatial features and class boundaries. While the Residual Channel-attention network (RCA) demonstrates significant advancements in high-resolution remote sensing image scene classification, it is essential to acknowledge certain limitations that emerged during our study. One notable consideration is the sensitivity of the RCA network to specific dataset characteristics, such as imbalances or variations in scene complexity. Additionally, computational efficiency, while addressed to a large extent, may still pose challenges when dealing with extremely large datasets. The RCA network's performance may be influenced by the choice of hyperparameters, and further optimization and sensitivity analysis are needed. These limitations warrant careful consideration when applying the RCA network in diverse scenarios. To pave the way for future research in remote sensing image classification, we identify several promising avenues for improvement and exploration:

- **Advanced Contextual Modeling Techniques**

Exploring sophisticated contextual modeling techniques, including graph convolutional networks and various attention mechanisms, can enhance the RCA's capability to capture

intricate relationships within complex scenes. Integrating these methods could further refine the network's performance across various datasets and scene complexities.

- **Domain Adaptation Strategies**

Addressing the challenge of domain shift in remote sensing images is critical. Our future investigation will delve into assessing the network's capacity to extract features that remain invariant despite domain shifts. This goal aims to ensure that the learned feature distribution aligns with diverse domains, thus giving the model the adaptability to varying data distributions. The investigation will specifically explore the potential integration of subspace alignment and instance weighting techniques for effective domain adaptation [49]. Subspace alignment methods aim to align the subspaces of source and target domains, facilitating the extraction of domain-invariant features. Concurrently, instance weighting approaches assign different weights to instances from the source and target domains, highlighting the significance of samples that contribute more to adaptation. By applying these strategies, the robustness and generalization ability of the proposed network will be enhanced across different domains [49].

- **Uncertainty Estimation**

Considering the importance of uncertainty estimation in remote sensing applications, future work could delve into techniques such as Bayesian neural networks or dropout regularization. This would provide a more comprehensive understanding of the RCA network's predictions and contribute to its robustness. Replacing or modifying some layers in our network to incorporate bayesian layers. These layers will model the uncertainty in the weights. Moreover, adjust the loss function to include a term that accounts for the uncertainty in predictions. This is often a combination of a traditional loss term and a regularization term related to uncertainty. Moreover, training the network using Bayesian training techniques. This might involve sampling from the weight distribution during training. We plan to enhance our proposed network by accounting for uncertainty in its predictions. One approach is to employ the Monte Carlo dropout method, where dropout layers are applied during both training and testing phases [50]. This involves predicting each new input multiple times (K times), and the average of these predictions determines the final class. Another approach is the use of a Bayesian framework, which allows the model to learn relationships between input data and different types of uncertainty. This includes aleatoric uncertainty, which captures inherent variability, and the approximation of epistemic uncertainty. By incorporating these methods, our goal is to make the proposed network more robust and reliable.

6 Conclusion

In conclusion, this work addresses the limitations of traditional convolutional networks in capturing complex semantic relationships from multi-scale ground objects in remote sensing scene images. To overcome these challenges, we introduced the Residual Channel-Attention (RCA) network, a novel and lightweight solution designed to enhance feature extraction and classification accuracy in remote sensing tasks. The RCA network's key innovations lie in its integration of residual connections and a channel-attention mechanism. The residual connections enable efficient multi-scale feature extraction, allowing the network to capture both fine and coarse details. Meanwhile, the channel-attention mechanism selectively emphasizes important feature channels while suppressing irrelevant background information, resulting in more refined feature representation and improved classification performance. This proposed

mechanism helps our network excel in handling the inter-class similarity and high intra-class variability.

Our experimental results, based on three benchmark datasets (RSSCN7, PatternNet, and EuroSAT), demonstrate the RCA network's superior performance, with classification accuracies of 97%, 99%, and 96%, respectively. These results show that the RCA network not only outperforms existing state-of-the-art methods but also significantly reduces computational complexity, particularly in challenging scenarios with limited training data and complex image backgrounds. Visualization using Grad-CAM further validated the effectiveness of our approach, confirming the network's ability to focus on the most critical image features. In light of these findings, we recommend extending the application of the RCA network to other remote sensing tasks, such as object detection and change detection. Additionally, future work could explore improving the network's robustness across different environments through domain adaptation techniques. Our RCA network provides a strong foundation for future advancements in remote sensing image classification, offering a versatile and efficient framework for handling increasingly complex datasets.

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Data Availability The datasets generated during and/or analyzed during the current work are available in the [Kaggle] repository, <https://www.kaggle.com/datasets/nifulislam/rsscn7-dataset> <https://paperswithcode.com/dataset/patternnet> <https://www.kaggle.com/datasets/apollo2506/eurosat-dataset>

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this manuscript.

Conflicts of Interest The authors declare that they have no conflict of interest.

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